

Embodied cognition and its significance for education

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journals.sagepub.com/home/tre**Lawrence Shapiro**

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Abstract

Psychology has made, and continues to make, a significant contribution to the discipline area of education. Since one of the main aims of education concerns student learning – which is an indisputably psychological phenomenon – we argue that the emerging research agenda of embodied cognition has much to offer educational practitioners, researchers, and/or policy-makers. Although embodied cognition is still in its infancy, the multidisciplinary and interdisciplinary nature of the literature provides some thought-provoking recommendations to enhance educational practice or practices, which in turn can bring about student learning more effectively. Consequently, this article will be concerned with the discussion of two issues: first, we provide a brief historical overview that foregrounds embodied cognition, and, second, we outline the educational implications of embodied cognition through the use of some examples significant to education. We conclude with an argument for the importance of making findings in an area we call ‘embodied education’ available to teachers.

Keywords

Cognition, education, educational research, embodied cognition, learning, psychology

Introduction

Most would agree that in a broad sense education involves *learning*, comprises *learners*, is associated with the intentional activity of *teaching*, and is closely linked with clearly demarcated spatiotemporal *educational settings*, such as schools and universities. Taking into consideration some psychologists’ interest in *learners* and *learning* from the early

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twentieth-century onward, particularly within educational contexts (e.g. Thorndike, 1910), it is not too hard to see how cognitivist or behavioral schools of thought have in turn influenced theories of learning in educational discourse and practice. In the former case, cognitivist views of learning are basically concerned with internal mental factors that influence cognition, such as how we organize and reorganize our thinking as a result of our experiences in the world (Piaget, 1960 [1926], 1950 [1947]), whereas in the latter case, behavioral accounts of learning are generally concerned with external factors, such as the identification of positive and negative reinforcements for certain types of behaviors (Skinner, 1965, 1968). The strengths and limitations of both these accounts are various; however, our intention at this juncture is to elucidate how each has failed to understand the role our embodiment (mind *and* body) plays in cognition.¹ Indeed, recent findings from research literature on learning and cognition from a diverse array of discipline areas, such as philosophy, psychology, linguistics, neuroscience, and computer science, have contributed to the view that traditional cognitivist accounts of the mind should be challenged because they exclude the close relationship that exists between mind and body that is more profound than initially considered (Shapiro, 2004, 2007, 2011, 2012). The central challenger emerging is the multidisciplinary and interdisciplinary research area known in the literature as ‘embodied cognition’ (e.g. Calvo and Gomila, 2008; Coello and Fischer, 2016; Fischer and Coello, 2016; Shapiro, 2004, 2007, 2008, 2011, 2012, 2014; Wilson, 2002).² Varela et al.’s (1991) book titled *The Embodied Mind* is commonly seen as the catalyst for the embodied cognition program because they argue that cognition is

... inextricably linked to histories that are *lived*, much like paths that exist only as they are laid down in walking [because] ... cognition is no longer seen as problem solving on the basis of representations; instead, cognition in its most encompassing sense consists in the enactment or bringing forth of a world by a viable history of structural coupling. (p. 205)

Such an account not only brings to our attention that cognition is grounded in our embodiment (embodied action) and that these histories are ‘*lived*’ (experiences) but also brings to our attention how adaptation to our environment (natural drift) has resulted in a cognitive system that is enacted through structural coupling and also constrained by the pathways laid down. Indeed, findings by Varela (1979, 1992, 1999; Thompson and Varela, 2001) and Maturana (Maturana and Varela, 1990, 1998) represent an ongoing commitment to understand the embodied processes of learning and have opened up new theoretical paths concerning learning and cognition that are less linear, hierarchical, and representational than earlier conceptions found in the literature. In *The Tree of Knowledge*, they highlight the ‘... inseparability between a particular way of being and how the world appears to us ...’ which is nicely summed up in the aphorism, ‘All doing is knowing, and all knowing is doing’ (Maturana and Varela, 1998: 26). In a sense, how we make distinctions that emerge in action and experience is central to learning, and hence why Maturana and Varela (1998) refer to the embodiment of learning as something that is ‘circular’ in nature. This circularity is evident in the capacity of human beings to reflect on ‘knowing how we know’ through the ‘act of turning back upon ourselves’ in a continuous process of both discovering our own ‘blindness’ and determining our ‘certainties’ (Maturana and

Varela, 1998: 24). This constant process of backwards-and-forwards in a circular feedback loop involving the mind and body in an environment leads to what Varela et al. (1991) refer to as the 'middle way' of learning between an objective and subjective certainty of experience. Understandably, embodied cognition is concerned with the mind, body, and the environment, and as a result promises to upend traditional ways of thinking about a number of long-studied topics, such as perception, language acquisition, social interaction, memory, and reasoning.³ Subsequently, for the purposes of this article, we will be concerned with the discussion of two issues: first, we provide a brief historical overview that foregrounds the emerging research area of embodied cognition, especially the important insights that distinguish embodied cognition from its computational forebear, and, second, we outline the educational implications of embodied cognition through the use of some examples significant to education and provide an argument for the importance of making these findings accessible to teachers in their educational practice or practices. It is important to note that while there presently exists some activity in what we might call 'embodied education', 'embodied learning', 'embodied perspective' or 'embodied perceiving', 'embodied grounding', and so on,⁴ we believe that the emerging interdisciplinary research agenda of embodied cognition contains fertile ground whose surface has, to date, merely been scratched. We hope this article goes some way toward deepening the furrows.

A brief historical overview of embodied cognition: Foregrounding an emerging research area

A convenient entry into embodied cognition begins with a simplified description of the conception of mind that it seeks to replace. This conception, at the core of the cognitive revolution that displaced behavioristic psychology in the middle of the last century, depicts psychological processes as computational, involving algorithmic operations over symbolic representations. In contrast to behaviorism, which concerned itself with the discovery of laws relating observable stimuli to observable behavior, traditional cognitive science focuses on the processes internal to an organism, that is, the unobservable (by direct means), and presumably computational, processes by which symbolically encoded stimuli are transformed into symbolically encoded instructions that result in the production of intelligent behavior. In short, cognitivist attempts to probe the internal workings of the 'black box' that the behaviorist regards as impenetrable to experimental investigation.

Following from this computational theory of mind (as it is now called) are three central commitments that embodied cognition challenges:

1. **Body-mind dualism:** Although cognitivists do not subscribe to anything like Cartesian dualism, according to which minds are non-physical substances, they do adhere to a dualism of a different sort. Insofar as psychological processes are computational, they can be investigated without regard to the 'hardware' in which they are implemented. Perhaps the most obvious way to see this point is to consider that the same software program, for example, PowerPoint, can be run on

different kinds of hardware, for example, Mac and PC computers. Just as one can investigate the properties of PowerPoint without considering the physical features of the computer chips and circuitry on which it is running, so too the psychologist can investigate minds without regard for their bodily implementation.

2. Amodal symbols: A symbol is amodal when the relationship between it and what it represents is arbitrary. For instance, the word 'martini' is a symbol that represents a particular kind of cocktail. But the connection between the word and the cocktail is arbitrary in the sense that there is no reason *that* word should be used to stand for *that* cocktail. 'Martini' might have been used to refer to a kind of marsupial. Nothing mandates that it be used to represent a cocktail. Similarly, the binary symbol for number seven is '111', whereas the more-standard Arabic symbol is '7'. But, as with 'martini', there is no reason that these symbols, rather than others, should be used to name number seven. Computers, obviously, encode information in amodal symbols. The '1's' and '0's' (or, more accurately, the positive and negative magnetic charges) with which they encode information are only arbitrarily connected to their referents. Correspondingly, if psychological processes are computational, then the connection between patterns of neural activity and what these patterns represent will also be arbitrary. Indeed, this is another reason why, as mentioned above, the psychologist need not care about implementational details as there is no intrinsic connection between the content of the mind and symbolic 'vehicles' that bear this content.
3. Poverty of the stimulus: Although this idea is not directly entailed by a commitment to computationalism, it has long been associated with it. Cognitive scientists have long held that the stimulation an organism encounters is impoverished – it underspecifies the world – and thus it is the computational brain's job to process this stimulation according to rules that 'fill in' this missing information. Chomsky, for instance, proposed that human beings are equipped with an internal set of linguistic rules, a grammar that makes sense of the messy stream of speech they encounter in their first years of life. Similarly, computational vision theorists (e.g. Marr, 1982) assume that information on the retina requires numerous stages of computational processing in order to represent accurately the layout of surfaces in the environment.

Interestingly, this commitment to the idea that cognition is an *ampliative* process, the job of which is to convert a sparse or impoverished representation of the world into something richer and more accurate, can underlie different conceptions of computation. For instance, the ampliative assumption guides not only traditional computational approaches to cognition but more contemporary approaches, such as those pursued by so-called 'connectionists', as well (Rumelhart and McClelland, 1986). To be sure, tremendous controversy surrounds questions about how to make sense of concepts such as representation, rule-instantiation, and computation within connectionist machines (Hatfield, 1991), but, whatever the outcome of these disputes, it is clear that connectionist machines do compute insofar as they transform inputs into outputs. Also clear is that connectionist models of cognition share with traditional computationalism the view that such computation is necessary in order to convert an initially impoverished stimuli (or

inputs) into something more detailed. Replacing the ‘assumptions’ that receive explicit representation in classical computational algorithms are excitatory or inhibitory connections between nodes that implicitly ‘contain’ assumptions about the world in the form of the weightings they acquire as the connectionist network is ‘trained up’. For this reason, connectionism, despite its differences with traditional ‘rules and representations’ analyses of computationalism, might still be sensibly construed as a kind of computationalism, the purpose of which is to embellish impoverished stimuli into something that can be of eventual use to a cognitive agent.

Let us now consider how embodied cognition responds to the cognitivist’s commitment to mind–body dualism, amodal symbols, and poverty of the stimulus. We noted that these commitments reflect an allegiance to a computational theory of mind. But embodied cognition has no such allegiance. Instead, the embodiment theorist’s guiding assumption is that an organism’s body is, in some sense, integrated in cognitive processing. Exactly how to understand this integration remains controversial, but examination of some research projects helps to clarify the idea.

A number of studies indicate that the body plays a role in language processing (Glenberg, 2008, 2010, 2015; Glenberg et al., 2013; Glenberg and Kaschak, 2002). If mind–body dualism of the sort to which computationalism is committed were true, then, some embodiment theorists have argued, we should not expect the body to influence language processing in the way that it appears to do. Glenberg et al. (2008) have performed a number of experiments that reveal a surprising connection between, on the one hand, a subject’s capacity to understand a sentence and, on the other, bodily actions that the subject is asked to make either prior to or during judgments of sentence sensibility. In one such experiment, subjects must move 600 beans from a wide-mouthed jar into a narrow-mouthed jar. In the ‘away’-condition, subjects moved the beans from the proximate wide-mouthed jar to the distal narrow-mouthed jar. The direction of motion is reversed in the ‘toward’-condition. Following this task, subjects are exposed to sentences that are either sensible or non-sensible. The critical sensible sentences describe an away- or toward-transfer (e.g. ‘You deal Mark the cards’; ‘Mark deals you the cards’). An example of a nonsense sentence is ‘Mark deals the cards you’. Surprisingly, subjects who engaged in the away-condition were slower to judge as sensible away-transfer sentences, and subjects engaged in the toward-condition were slower to judge as sensible toward-transfer sentences.

Before speculating about why repetitive motion in a particular direction slows comprehension of transfer sentences in the ‘same’ direction, we should pause to note how odd these results are from the perspective of computational cognitive science. If language processing were purely computational, then why should the movement of beans make a difference to judgments of sentence sensibility? Moreover, Glenberg et al. (2008) found that when subjects moved beans with their right hand and were asked to register a sensibility judgment on a keyboard using their left hand, the effect disappeared. Why should a computational process show sensitivity not only to repeated arm motions but also to *which* arm is moving? No matter how one interprets these results, they would not be predicted on the assumption that language processes are computational.

So far, this experiment, and many others like it, implicates an unexpected influence of the body on language comprehension, thus calling into question the body–mind dualism

that follows quite naturally from the computational theory of mind. But, additionally, among Glenberg et al.'s (2008) explanations for the results is one that calls into question the computationalist's commitment to amodal symbols. Perhaps, Glenberg et al. (2008) suggest that the comprehension of sentences about transfer actions involves a simulation of the action itself. That is, when reading the sentence 'You deal Mark the cards', a subject simulates a performance of this action. In this context, 'simulation' refers to the activation of some areas of the brain that would be involved were the subject actually to perform the action described in the sentence. Hence, when a subject reads a sentence about dealing cards to Mark, areas of the subject's brain that would be involved in such an action become active. Moreover, if the subject had been in the away-condition and then reads a sentence describing an away-transfer, the fatigue induced in the motor areas of the brain by the repeated away-motions as the subject moved beans away from himself inhibits or interferes with comprehension of the away-sentence.

Importantly, this explanation of the data denies the amodal nature of cognitive representations. The amodality of a symbol derives from the arbitrariness of its connection to the content it represents. As we observed earlier, 'martini' could as well represent a marsupial as a cocktail. The neural 'symbols' to which computationalists assign meaning are similarly arbitrary, in the sense that their format invokes nothing 'reminiscent' of the stimulus that causes their activation. But Glenberg et al.'s (2008) explanation of the results depends on the idea that when a subject reads a sentence, it is encoded in those areas of the brain that are involved in performing the actions the sentence describes. Thus, sentences about away- or toward-transfers are not encoded in symbols that bear only an arbitrary connection to the actions they represent. Rather, they are encoded in precisely those areas of the brain that are involved in performing the actions that the sentences describe (see also Barsalou, 1999). Instead of being formatted in an area of the brain that bears no connection to the experiences a particular stimulus elicits, embodied or modal symbols are tied directly to such experiences because they replicate the patterns of neural activation that occur in the brain when encountering the actual stimulus.⁵

Glenberg et al.'s (2008) explanation of these results is in keeping with a wealth of neuroscientific research, inspired by studies like this, that suggest a tight connection between cognitive processing and brain areas associated with physical motion. Thus, we now know that reading a sentence about kicking a ball will activate areas of the brain that are involved when actually kicking a ball (Pulvermüller, 2005). We know that application of transcranial magnetic stimulation to areas of the brain involved in motor control will interfere with comprehension of sentences that describe actions (Pulvermüller, 2005). We also know that understanding others' actions requires activation of those parts of the brain that would be involved when taking similar actions, oneself (Gallese et al., 1996; Rizzolatti and Craighero, 2004). These neuroscientific findings all lend support to the idea that the brain represents the world not through amodal symbols but, in effect, by trying to imagine a body in active engagement with the world.

Turning finally to the computationalist's assumption regarding the poverty of the stimuli to which organisms respond, we find embodiment theorists embracing the ecological psychologist J. J. Gibson's idea that, in fact, the stimuli an organism encounters contain all the information necessary for perception (Gibson, 1950, 1966). Computationalists miss this fact because, Gibson held, they do not appreciate how

embodied organisms, in active engagement with their environments, can 'pick up' the very information that they believe to be absent. According to Gibson, an organism's movement through the environment will reveal certain invariances in the stimulus array it confronts, rendering unnecessary a need for the computational embellishments to information that traditional cognitive scientists take for granted.

As an example of this idea, consider the extensively studied question about how an outfielder standing more than 200 ft from home plate is able to situate himself precisely where he needs to be to catch a fly ball. A computational solution requires that the brain processes information about the velocity of the ball following its impact with a bat, its direction, and its angle. The brain then, presumably, applies sophisticated calculations to deduce the location where the ball will drop. But the alternative embodied approach requires only that the outfielder move laterally until the ball no longer appears to be moving along a curved path, but instead appears to be rising in a straight line. This pattern of motion carries the information that the outfielder is now standing precisely where he needs to be to catch the ball (McBeath et al., 1995).

So far, we have presented some of the main ideas that distinguish embodied cognition from computational cognitive science. Together, the ideas suggest a new way of approaching old topics. For instance, decades of standard practice in a perceptual psychology laboratory would have subjects sitting in front of a computer display making judgments about stimuli as they flash across the screen; however, if the perception is embodied, such a practice prevents the very sort of active, environmental engagement on which cognition depends.

Similarly, long-standing questions about certain kinds of behavior might, from the embodied perspective, have been looking for answers in the wrong places. As an example, psychologists have puzzled over why 7- to 12-month-old infants will make a strange sort of reaching error. Having observed and retrieved an attractive toy placed under one cup, the infant then sees the toy moved beneath another cup, but the infant will continue to reach for the first cup. A traditional explanation attributes the error to a mismatch between the object representation the infant has internalized and an ability to place reaching behavior under the control of this concept (see Piaget, 1955 [1937]). But the embodiment explanation eschews talk of representations in favor of an account that focuses on relations between the infant's body – the mass of its arms – and variables such as the distance between the infant and the toy, as well as the delay between hiding the object and reaching for it. This second explanation trades in the computational equipment associated with reference to things like object representations, for the equipment of dynamical systems theory, which is well suited to examine the interactions between variables such as arm mass, distance of stimulus from the infant, and distance between the two cups. Such an explanation has now made predictions and answered questions about an infant's reaching behavior that first kind of explanation has missed (Thelen et al., 2001).

Similarly, behavior that appears inexplicable or completely unexpected from a computational perspective lends itself to an embodied treatment. Casasanto (2009), for instance, told subjects a story about a child who loves zebras and thinks that they are good but hates pandas and thinks that they are bad. Subjects were then shown two boxes in a row and were asked to draw the animals in the boxes where they belonged. Right-handed subjects drew the good zebra in the box to the right and the bad panda in the box

to the left; left-handed subjects did the reverse. Like in the experiment of Glenberg's we discussed above, Casasanto's explanation for this strange behavior assumes that subjects' concepts of good and bad are represented *modally*: the concept of good in a right-handed person is in part represented in areas of motor cortex in the left hemisphere (which controls the right hand), and this is because right-handed subjects prefer (and so develop good, rather than bad, associations with) actions to their right.

Given the tremendous variety of issues to which an embodied perspective might be usefully applied, the natural question seems not to be *whether* embodied cognition might help to inform educational practices, but *how*. As a result, in the next section of this article, we offer some examples relevant to education and argue that teachers ought to be incorporating findings from the field of embodied education into their educational practice or practices.

The educational implications of embodied cognition: Some examples significant to education

We are cognizant of recent theoretical and empirically oriented literature of embodied cognition in educational contexts, particularly from a science, technology, engineering, and mathematics (STEM) point of view (e.g. Abrahamson and Lindgren, 2014; Abrahamson and Sánchez-García, 2016; Hutto et al., 2015; Hutto and Sánchez-García, 2015; Lakoff and Núñez, 2000; Nathan and Walkington, 2017; Newcombe and Weisberg, 2017; Núñez et al., 1999; Pouw et al., 2014; Radford, 2003, 2009). As such, our intentions are to extend on this literature as a means to demonstrate the educational implications of embodied cognition. Take, for instance, the hypothetical classroom scenario. A science teacher wants to introduce an unfamiliar instrument, such as two bicycle wheels that can spin independently on a single axle (similar to gyroscope wheel, but with two wheels), to his or her class for the first time. From the range of possible options available to the teacher, he or she could (1) describe the properties of the instrument with an emphasis on particulars like its shape, sound, color, and so on; (2) show a video that demonstrates the physical properties of the instrument and how it is used; and/or (3) take students to a science laboratory to observe an experiment being conducted with the instrument while providing an opportunity to experiment with the instrument themselves. Of course, there are other possible methods that could be employed to teach the concepts of torque and angular momentum in physics. However, our point is to highlight how physical experience can enhance and positively influence students' learning of scientific concepts (e.g. Konya et al., 2015). Understandably, some teachers may want to know the reason or reasons why physical experience can enhance learning through instructional manipulatives. Indeed, embodied cognition is concerned with the interaction of the mind, body, and environment in explaining how knowledge is grounded in sensorimotor routines and experiences (Barsalou, 1999, 2008; Lakoff and Johnson, 1999). As a result, in this section, our attention turns to the educational implications of embodied cognition through some specific examples relevant to education.

An interesting starting point is a popular view of teaching and learning that argues mastery of discipline-specific knowledge should take place first (formalisms first view of learning), before it can be applied (Nathan, 2012). Such a view has been found to be

deeply questionable because it tends to reinforce a formalisms-only mind-set toward learning and teaching that is rooted in dualistic views of knowledge that fallaciously associate intellectual work with the ‘mind’ and practical work with the ‘body’—precisely the distinction that embodied cognition denies. Indeed, the research literature highlights some of the problems with viewing learning as a change (only in one direction) in representations from concrete to abstract or abstract to concrete (so-called ‘transfer of learning’) because instructional manipulations of learning do not necessarily involve a linear type of shift (Pouw et al., 2014). In this case, embodied cognition would claim that learning is not dependent upon the establishment of a complex set of symbolic rules that are decontextualized from sensorimotor experiences and the environment. In contrast, from the perspective of embodied cognition, learning is contingent upon the cognitive activity that is triggered by the environment and is determined by the dynamic nature of living beings engaged in the self-organizing activities by which they sustain themselves (Maturana and Varela, 1998). This development marks a gradual transition that involves a form of embodied knowledge, where the learner either becomes less dependent on external support over time or learns what kind of external support can be used in place of more costly ‘internal’ cognitive machinery, as in the case of the fly ball. Although this transition or ‘structural drift’ is due to the learners’ interaction with an environment, the environmental parameters fix the boundaries where learning can be promoted or restrained; however, it is important to note that this does not determine the direction or outcome of learning as this is structurally determined (Horn and Wilburn, 2005). Embodied cognition supports the differing view that learning involves synchronous cycles of activity and the dynamic balance of support that emerges from interactions with the environment. For instance, when an outfielder in baseball catches a fly ball, it may appear that they are dependent on sophisticated cognitive operations, when in fact they are exploiting features of the environment in a way that reduces cognitive load.

Schwartz et al. (2005) highlight what they consider to be a problematic division in the literature concerning the learning of firsthand (direct experience) and secondhand knowledge (descriptions of experience). They argue that because most formal learning is associated with the application of secondhand knowledge (often through language), it is problematic because it requires that students interpret descriptions of the world in the absence of the original referent. Similarly, firsthand learning in isolation is equally problematic because direct experiences need to be coupled with secondhand knowledge to be given any meaning. Quite rightly, Schwartz et al. (2005) argue that when it comes to learning, the focus should not be on ‘either/or’ in an attempt to isolate certain learning outcomes, but more of an emphasis on ‘both/and’ as a means to integrate firsthand and secondhand knowledge. In fact, research by Glenberg (2008, 2010) supports this integrative approach to learning and the importance of both physical manipulation and imagined manipulation in mathematics and reading comprehension. In particular, it is worth noting the importance of physical manipulation before imagined manipulation. The reason for this appears to be due to the significance of the practical manipulative experience grounding abstract symbols (i.e. words and mathematical symbols) in the internalization of embodied mental models (Glenberg, 2008, 2010).

One area where embodied cognition research has interesting educational implications concerns the role gestures play in learning. For instance, research by Goldin-Meadow

(Church and Goldin-Meadow, 1986; Cook et al., 2005; Goldin-Meadow, 2003, 2009; Goldin-Meadow and Butcher, 2003; Goldin-Meadow et al., 1993, 1999, 2001, 2007, 2009; Goldin-Meadow and Singer, 2003; Goldin-Meadow and Wagner, 2005; Novack and Goldin-Meadow, 2015; Perry et al., 1988) and her colleagues have investigated why people gesture when they speak. Curiously, such gestures occur even when the conversational partner is on the other end of a telephone call. Not only do we gesture as we speak to people who may be miles away from us, but blind people who have never seen a gesture also gesture when they speak. Why is speech so closely associated with gesture? Does it convey information not found in speech?

One hypothesis receiving increasing support is that gesture is itself a form of communication that can either duplicate the information conveyed in speech or, more interestingly, contribute information that diverges from that present in speech. Indeed, Church and Goldin-Meadow (1986), Goldin-Meadow and Wagner (2005), and Perry et al. (1988) have examined the relationship between gesture and speech used by children. Church and Goldin-Meadow (1986) and Goldin-Meadow and Wagner (2005) observed children who were asked whether the quantity of water in two tall glasses was the same and then whether the quantity of water remained the same when the contents of one of the glasses was poured into a low, wide glass. Many children were *non-conservers*: that is, they judged the quantities to be the same when the water reached the same height in the two tall glasses, but different when the height of the water was lower in the wide glass. Some non-conservers would explain their judgment by *saying* that the height of the water in the two glasses differed and by *pointing* to the height of the water in the tall glass and then pointing to the height of the water in the wide glass. In these cases, the gestures and the speech conveyed the same information about water height. Other non-conservers, however, displayed a mismatch between their speech and gestures. This gesture–speech ‘mismatch’ was first identified by Church and Goldin-Meadow (1986) in their study of children’s explanations of conservation problems. They found that the combination of gesture and speech can provide an insight into transitional knowledge states of children, particularly as they grapple to make connections between old knowledge states and new concepts. Indeed, they found that a subset of children tended to use gestures that differed in content from what was conveyed simultaneously in speech. To understand this phenomenon, Church and Goldin-Meadow (1986) introduced two new terms: ‘discordance’, which refers to explanations in which gesture fails to match speech, and ‘concordance’, which refers to explanations in which gesture and speech convey the same content. Church and Goldin-Meadow (1986) concluded that gesture–speech measurement of discordance provides a useful tool in identifying a child’s level of understanding of a concept, and hence readiness to learn. As such, subsequent studies by Perry et al. (1988) highlighted how different types of gesture–speech discordance with respect to a particular concept could be used by teachers to tailor instruction to the child’s current level of understanding.

From an educational point of view, what is interesting about this finding relates to the way children who display discordance are more receptive to instruction about conservation than children who display concordance (Goldin-Meadow, 2009; Goldin-Meadow and Singer, 2003; Goldin-Meadow and Wagner, 2005; Perry et al., 1988). This led Goldin-Meadow (2009) to conclude that discordance is a sign of readiness to learn because it indicates the presence of an idea that is not quite available to the subject’s

conscious awareness and so is not quite ready to be articulated in speech. It is as if part of the subject – the part that speaks – believes that the quantities of water differ, but another part – the part that gestures – understands that differences in the width of the glasses are relevant to conservation judgments. Unlike the subjects who exhibit concordance, those whose gestures and speech show a mismatch are closing in on the correct understanding of conservation and thus primed for further educational interventions by the teacher to elicit dynamic gesture production that corresponds to their spoken explanation. Indeed, it was found that when teachers are aware of gesture, they pick up information about their students' cognitive state that was not available in their speech (Goldin-Meadow, 2009; Goldin-Meadow and Singer, 1999, 2003; Novack and Goldin-Meadow, 2015). Observation of teachers revealed that they adopt different teaching strategies for students who display either discordance or concordance, relying more on mismatches of their own when explaining principles of conservation to students who mismatch (Goldin-Meadow and Singer, 2003). Similarly, the same study by Goldin-Meadow and Singer (2003) showed that teachers who display discordance benefit student learners because they expose them to a range of strategies in the gestural modality, thus teaching them how to 'think with gestures' about certain problems.

Additional work on gesture reveals its importance in the acquisition of mathematical concepts. Alibali and Nathan (2012), following McNeill (1992), distinguish between different kinds of gestures. Pointing gestures draw attention to objects, individuals, or locations; iconic gestures, typically created by shaping the hands in particular ways or by 'drawing' in the air, represent objects (e.g. a bowl) or shapes (e.g. a triangle); and metaphoric gestures convey meaning through metaphoric extension, as, for instance, when rocking one's hand back and forth to indicate that one is 'weighing' an idea. Alibali and Nathan describe how teachers might exploit each kind of gesture to clarify, for example, the order of operations in an equation, or present the relationships between analogous geometrical shapes, or explain how to conceive of a line's slope. In these cases, the body becomes a conveyer of information that might be used to supplement or replace the information provided by symbolic constructions of the sort more standardly associated with educational instruction, that is, words or writing on a board. In addition, the use of gesture provides students with a new tool with which to express or try out ideas. Insofar as this new tool does the work that would otherwise fall to 'pure mentation', cognitive load is reduced (Roth, 2001), in a way analogous to that in the outfielder case we described earlier.

Drawing from the studies of gesture, we would like to state concisely four educational implications. The first three are specific to the role of gesture in learning. The last is more general and speaks broadly to the importance of embodiment in instructional contexts. We close this section with a more detailed examination of this fourth implication:

1. Teachers could and should look for concrete cues such as gesture–speech mismatches in order to identify students who have not fully comprehended the concept being taught. In response to these mismatches, teachers could increase the proportion of gesture–speech matches they use in teacher instruction, particularly when instructing students who are in transitional knowledge states.
2. The use of gesture in teacher instruction encourages learners to produce gestures of their own, or imitate the gestures that their teachers produce, which can

enhance learning. In addition, encouraging students to gesture allows knowledge to be conveyed through their bodies that cannot verbally be communicated, but most importantly it demonstrates that the student is ready to learn.

3. Gestures can be classified into different categories, with each category defined by a particular function. For instance, gesturing is known to either alter the learners' responses and thoughts or lighten the cognitive load of the teacher or learner because it shifts the cognitive load from verbal to visuospatial stores, thus permitting the individual to work harder on the task and/or change their representation of the task in a manner that facilitates learning. Teachers who acquaint themselves with the distinct purposes of different kinds of gesture will be able 'read' and communicate more effectively with their students.
4. Embodiment offers either a causal route to more effective learning or a diagnostic tool for measuring conceptual understanding, and thus, educational 'best practice or practices' require that instructors keep abreast of current research in embodied education.

As indicated above, this fourth implication is more general than the preceding three and also, due to its prescriptive nature, perhaps most in need of additional elaboration and defense. Of course, the preceding discussion of gesture anticipates this fourth point to an extent; however, the research we examined risks leaving the impression that gesture develops spontaneously and that the effective instructor's job consists mainly in learning the meanings that different gestures communicate and, perhaps, in modeling gesture for the purpose of facilitating students' acquisition of gesture. Doubtless, a teacher who masters both these tasks will be more effective, *ceteris paribus*, than one who does not. Yet, the point we wish to emphasize now is this: occasionally the gestures and, more generally, bodily actions that students exhibit spontaneously when learning a task are *wrong for the task*, in the sense that they diminish rather than improve task performance, or they are diagnostic of conceptual misunderstandings and thus useful in assessing a student's comprehension of a topic.

This claim, that some embodied learning strategies are in fact detrimental to learning or indicative of conceptual misunderstanding, constitutes the first premise in our argument for the normative conclusion that we draw in (4) above. The second premise is simply that which gestures and other bodily actions correlate with success in learning and which indicate conceptual deficiency are a matter for empirical investigation. But, third, such empirical investigation is a primary focus of research in the field of embodied education. Hence, we conclude, educational 'best practice or practices' require that teachers remain informed of research findings from the field of embodied education.

Before turning to our defense of the premises in this argument, we offer a couple of preliminary remarks. First, we began this article with the observation that psychology and education are necessarily intertwined, given that the purpose of education is to promote learning and that learning is a psychological capacity. We take the argument we are presently developing to move beyond this truism and to establish in more forceful terms the significance of a particular psychological research program – embodied cognition – to the educational mission. Second, we recognize that the argument that we sketched in the paragraph above is not, formally speaking, deductively valid. This is because fields

other than what we have been calling *embodied education* could, conceivably, shine light on which embodied learning strategies are to be preferred, and so, perhaps, teachers might look elsewhere for this kind of information. We accept this point but submit that, presently, the field of embodied education appears to be the best source of information about how bodily actions might contribute to or provide information about learning, and so rest content that our argument, even if not deductively valid, provides a compelling case for our recommendation that educators avail themselves of the findings that emerge from embodiment researchers.

Clearly, it is the first premise in our argument that requires most justification, and we shall do so by considering two studies. Both studies reveal that the gestures a student adopts when solving a particular problem may be either poorly suited to the task or indicative of conceptual confusion. In either case, a teacher who is unfamiliar with the relevant literature on embodied cognition will miss an opportunity either to correct the student, offering her or him a more effective means of ‘embodying’ a solution, or to recognize that the student’s gestures are a symptom of conceptual confusion.

Before turning to these studies, we offer a word of caution with respect to their interpretation. Both studies reveal a correlation between a style of gesture and extent of success in task performance. However, neither study, in our view, establishes whether the style of gesture one adopts *causes* one to be more or less successful in task performance or, in contrast, whether one’s understanding of the task determines how one chooses to gesture. This indeterminacy lies behind our decision to formulate our first premise as a disjunction. If style of gesturing makes a causal difference to how well one performs on a given task, clearly educators should teach and encourage the use of effective gesturing. On the contrary, if poor performance is merely correlated with rather than caused by the choice to use a particular style of gesturing, then an educator has the opportunity to regard such gestures as a diagnostic symptom – an indication that a student may be conceptually deficient in some respect. As we describe the studies below, we will retain this attitude of neutrality toward the question of whether gestures are causes or indicators of conceptual understanding. That is, we will take the studies to show that choice in gesture makes either a causal difference to task performance or that it marks a student as possessing a particular level of conceptual clarity.

The first study (Walkington et al., 2014) involves college students who are asked to prove a mathematical conjecture: the sum of the lengths of two sides of a triangle is always greater than the length of the third side. Students were broken into two groups. Those in the control group were seated in front of the computer screen on which the conjecture was displayed and given pen and paper; those in the experimental group stood in front of the computer screen and were not provided with pen and paper.

Students availed themselves of four distinct strategies when attempting proofs of the conjecture. Those who neither gestured nor used pen and paper were least successful (only 11.5% could provide a correct proof), and those who used pen and paper were slightly more successful (27.3%).⁶ Of special interest for present purposes, however, is the difference in success between students who used one or the other of two gesturing strategies. Those who produced ‘static depictive gestures’, in which hands were used to create a static representation of a triangle, akin to a figure drawn on paper, correctly proved the conjecture 34.3% of the time. In contrast, students who engaged in ‘dynamic

depictive gestures' offered a correct proof at a rate of 63.6%. As Walkington (2014) describes these gestures,

problem-solvers first represent an object, and then engage in fluid transformations of that object using the affordances of their body. For example, a problem-solver might 'collapse' the triangle formed with their hands into two line segments on top of each other. (p. 480)

The lesson we take from Walkington's study is that choice in gesturing strategy correlates with success in task performance. As mentioned earlier, this correlation may be evidence for a causal relationship, in which case students who are encouraged to gesture dynamically rather than statically will, in certain conceptual domains, be more likely to discover important insights. Alternatively, students who display static gestures may be signaling a conceptual misunderstanding – one that a shift to dynamical gesturing may help to alleviate or one that should be remedied in some other way.

We draw a similar lesson from an examination of Gerofsky's study of students' understanding of Cartesian graphs (Gerofsky, 2011). Gerofsky elicited gestures from a group of 8th grade (age 13) and 11th grade (age 16) students. The students, selected by their teachers, were representative of 'top', 'average', or 'struggling' mathematics students. This information was held from Gerofsky until the conclusion of her study. The students' task was simply to describe a variety of graphs using gestures, vocal sounds, and words, but to refrain from formal mathematical descriptions.

Gerofsky (2011) found that students could be fitted into one of three categories. Some produced minimal gestures: 'small movements of a finger, hand and arm, without a great deal of larger kinesthetic movement involving the spine' (p. 251). Interestingly, these gestures appear comparable to the static depictive gestures that Walkington et al. (2014) observed in her subjects, insofar as students seemed to imagine a static depiction of a graph in their mind's eye: 'it was as if they were tracing a small graph on a vertical plane of glass or a sheet of paper in front of their upper-body, using a finger-tip "pencil" ...' (Gerofsky, 2011: 251).

Similarly, students in the second category adopted gestures analogous to Walkington's dynamic depictive gestures: 'These gestured graphs involved noticeable movement of the spine and often markedly kinesthetic, whole-body movements. Some students' gestures required them to reach, move off balance or take a step or two' (Gerofsky, 2011: 251). The final category of students generated gestures that 'did not correspond accurately to the shapes of the graphs, and often large sections of the graph were omitted' (Gerofsky, 2011: 251).

In light of our earlier presentation of Walkington's study, it should come as no surprise that of the three gesturing categories, those who engaged in the most dynamic form were also top mathematics students, and those who exhibited static gestures were average. Students whose gestures were erratic and inaccurate were struggling in mathematics. We take these results as further confirmation of our claim that how a student gestures correlates with how well they will perform on a particular conceptual task. Dynamic gestures prove more effective in creating an understanding of graphs, or they correlate with more advanced understanding, whereas static gestures decrease task performance or indicate a student's conceptual deficiency.

We regard studies like the two described above as sufficient to justify the first premise in our argument: some embodied learning strategies are in fact detrimental to learning or indicative of conceptual misunderstanding. The remaining premises seem to us to require no special defense. Premise 2 states simply that the connection between style of embodied action and task performance is not something a teacher could know *a priori* – it is an empirical matter and thus a proper subject for experimental investigation. Premise 3 merely affirms that the embodied cognition research program is, presently, the best source of information about these connections between actions and conceptual learning. The conclusion we draw from these premises is at once modest but significant. Put modestly, it asserts just that teachers would be wise to attend to research in embodied cognition so that they can be aware of the causal or diagnostic value of their students' embodied actions. The significance of this conclusion, however, grows when we weigh the relative novelty of research in the field of embodied education (much of this research began only after the turn of the century) against its obvious relevance to educational practice or practices. As we hope our discussion of a variety of psychological studies revealed, the body is extensively integrated within learning processes. However, the 'newness' of embodied cognition research means that much of this information is yet to be disseminated; indeed, much of this information is yet to be discovered. Thus, we believe, our conclusion that teachers apprise themselves of embodied educational findings ought best be construed as a challenge and a clarion call. The research, we hope to have shown, has obvious significance for education, but its novelty and, in the eyes of more traditionally inclined psychologists, upstart status leave uncertain the accessibility of its findings to the very educators who are best situated to exploit them.

Above we saw various ways in which the body may contribute to cognition. It may influence cognitive processing through action, as it does in Glenberg's bean transfer experiment or, in a slightly different way, in an infant's failure to reach in the right place for a toy. It might make a difference to how thought and concepts are encoded, as we saw in Casasanto's work with right- and left-handers' judgments about good and bad. It might help to simplify an otherwise demanding task, as it does when relying on its motion relative to a fly ball in baseball in order to determine the right place to stand. Within an educational context, focus on embodiment encourages a teacher to look for bodily cues that might indicate the preparedness of a student to learn a new concept or to rely on bodily action to introduce or clarify mathematical concepts. With a properly structured environment, a student's entire body might be transformed into something else, for example, an asteroid, whose motions, because they are also the student's motions, become more easily tractable.

Conclusion

Education has turned to psychology to understand human learning and to guide best pedagogical practices. In this case, we argue that the emerging research agenda of embodied cognition has much to offer educational practitioners, researchers, and/or policy-makers. Although embodied cognition is still in its infancy, the multidisciplinary and interdisciplinary nature of the literature provides some thought-provoking recommendations to enhance educational practice or practices, which in turn can maximize the effectiveness of the

teacher in bringing about student learning. Even though our account of embodied cognition has been limited due to space restrictions, we would argue that there is considerable potential for further research and enough existing literature to suggest new ways to think about instruction and classroom design. This article provided a brief historical overview of embodied cognition, especially the important insights that distinguish embodied cognition from its computational forebear. We then turned our attention to the educational implications of embodied cognition through the use of some examples significant to education, showing not only how these findings could be used by teachers in their educational practice or practices but additionally why teachers ought to pay attention to them.

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Notes

1. Although we are fully aware of the ongoing philosophical debate surrounding the mind–body problem, for the purposes of this article, we use the concept of ‘mind’ in a broad psychological sense that characterizes a human being’s capacity to think, perceive, and feel their experiences in the world.
2. It is important to point out that there is no universal definition of the concept of embodied cognition found in the literature. This should not be viewed as something problematic, but more a case of reflecting the diverse array of disciplines in which it has emerged. Indeed, it could be argued that embodied cognition is located at the intersection between psychology (cognitive and behavioral psychology), science (cognitive and neuroscience), and philosophy (philosophy of mind). In fact, we would argue that the multidisciplinary and interdisciplinary nature of embodied cognition has the potential to enhance educational practice or practices, and in turn influence student learning.
3. The edited book by Shapiro (2014) titled *The Routledge Handbook of Embodied Cognition* gives a taste of the wide domain of issues that are now being investigated from an embodied perspective. For some other examples, see the following: *Joint Ventures: Mindreading, Mirroring, and Embodied Cognition* (Goldman, 2013), *Mind in Action: Experience and Embodied Cognition in Pragmatism* (Määttänen, 2015), *Perceptual and Emotional Embodiment: Foundations of Embodied Cognition, Volume 1* (Coello and Fischer, 2016), and *Conceptual and Interactive Embodiment: Foundations of Embodied Cognition, Volume 2* (Fischer and Coello, 2016).
4. See the following: ‘Embodied learning’ (Stolz, 2015), ‘The embodiment of learning’ (Horn and Wilburn, 2005), *Embodiment and Education* (O’Loughlin, 2006), ‘Embodied persuasion’

- (Briñol and Petty, 2008), ‘The role of embodied change in perceiving and processing facial expressions of others’ (Briñol et al., 2010), *Embodied Grounding* (Semin and Smith, 2008), ‘Embodiment for education’ and ‘Embodiment as a unifying perspective for psychology’ (Glenberg, 2008, 2010), and ‘An embedded and embodied cognition review of instructional manipulations’ (Pouw et al., 2014). For some recent examples of literature that applies embodied theories within an educational context, particularly learning, see the following: ‘Embodied cognition and STEM learning’, special issue in *Cognitive Research: Principles and Implication* (Newcombe and Weisberg, 2017), and ‘Mind your body: The essential role of body movements in children’s learning’, special issue in *Educational Psychology Review* (Chandler and Tricot, 2015).
5. This pattern of neural activation is often described as ‘offline’ to indicate that it does not cause the behavior that encounters with an actual stimulus would. The activation is better characterized as a simulation of what one would do were one confronted with the actual stimulus.
 6. We take this as positive evidence that there is indeed a causal relationship between gesture and comprehension, at least insofar as gesturing improves performance as compared to non-gesturing. However, we would like more evidence before inferring that one style of gesture causes better performance than another.

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